

Obstruction Theory and Characteristic Classes

A Brief Introduction to Obstruction Theory

In homotopy theory, one often encounters the following problem:

Given a pair of spaces (X, A) and a map $f : A \rightarrow Y$. Can f be extended to a map $F : X \rightarrow Y$?

Obstruction theory provides a systematic way to study this problem.

Toy Model : A map $f : S^n \rightarrow Y$ can be extended to a map $F : D^{n+1} \rightarrow Y$ if and only if f is null-homotopic.

Now we consider the general case. Let K be a CW complex, Y be a simple space, i.e., $\pi_1(Y)$ acts trivially on $\pi_n(Y)$ for all n . Under this condition, we can unambiguously use the notation $\pi_n(Y) = [S^n, Y]$. Suppose we have a map $f : K^{(n)} \rightarrow Y$, where $K^{(n)}$ is the n -skeleton of K . We want to extend f to a map $F : K^{(n+1)} \rightarrow Y$. To do this, we only need to extend f to each $(n+1)$ -cell e_α^{n+1} .

Suppose the attaching map of e_α^n is $\varphi_\alpha : S^n \rightarrow K^{(n)}$. Then $f \circ \varphi_\alpha$ is a map from S^n to Y . By the toy model, $f \circ \varphi_\alpha$ can be extended to D^{n+1} if and only if $[f \circ \varphi_\alpha] = 0$ in $\pi_n(Y)$. We in fact get a cochain $c^{n+1}(f) \in C_{cell}^{n+1}(K; \pi_n(Y))$:

$$\begin{aligned} c^{n+1}(f) : C_{n+1}^{cell}(K) &\rightarrow \pi_n(Y) \\ e_\alpha^{n+1} &\mapsto [f \circ \varphi_\alpha]. \end{aligned}$$

We call $c^{n+1}(f)$ the $(n+1)$ -**th obstruction cochain** of f . There are some basic facts about $c^{n+1}(f)$:

- f can be extended to $K^{(n+1)}$ if and only if $c^{n+1}(f) = 0$;
- If $f, f' : K^{(n)} \rightarrow Y$ are homotopic, then $c^{n+1}(f) = c^{n+1}(f')$;
- (Naturality) Suppose L is another CW complex and $\phi : L \rightarrow K$ is a cellular map, then $c^{n+1}(f \circ \phi) = \phi^\# c^{n+1}(f)$.

At the following, we give some important properties of $c^{n+1}(f)$ without proof.

Theorem 1. $c^{n+1}(f)$ is a cocycle, i.e., $\delta c^{n+1}(f) = 0$. Thus it defines a cohomology class $[c^{n+1}(f)] \in H^{n+1}(K; \pi_n(Y))$. We call $[c^{n+1}(f)]$ the $(n+1)$ -**th obstruction class** of f .

A similar concept is defined for homotopies. Suppose we have two maps $f_0, f_1 : K^{(n)} \rightarrow Y$ and a homotopy $h : K^{(n-1)} \times I \rightarrow Y$ between $f_0|_{K^{(n-1)}}$ and $f_1|_{K^{(n-1)}}$. We want to extend h to a homotopy $H : K^{(n)} \times I \rightarrow Y$ between f_0 and f_1 . To do this, we only need to extend h to each n -cell e_α^n . Let $\varphi_\alpha : S^{n-1} \rightarrow K^{(n-1)}$ be the attaching map of e_α^n , $\psi_\alpha : D^n \rightarrow K^{(n)}$ be the characteristic map of e_α^n . Then we can form a map

$$\begin{aligned} h_\alpha : (D^n \times \{0,1\}) \cup (\partial D^n \times I) &\rightarrow Y \\ (x, 0) &\mapsto f_0 \circ \psi_\alpha(x) \\ (x, 1) &\mapsto f_1 \circ \psi_\alpha(x) \\ (x, t) &\mapsto h \circ (\varphi_\alpha(x), t). \end{aligned}$$

Since $(D^n \times \{0,1\}) \cup (\partial D^n \times I) \cong S^n$, h_α defines a map from S^n to Y . It can extend to $D^n \times I \cong D^{n+1}$ if and only if $[h_\alpha] = 0$ in $\pi_n(Y)$. We in fact get a cochain $c^n(f_0, f_1, h) \in C_{cell}^n(K; \pi_n(Y))$:

$$\begin{aligned} c^n(f_0, f_1, h) : C_n^{cell}(K) &\rightarrow \pi_n(Y) \\ e_\alpha^n &\mapsto [h_\alpha]. \end{aligned}$$

We call $c^n(f_0, f_1, h)$ the **n -th deformation cochain** of (f_0, f_1, h) . If $f_0|_{K^{(n-1)}} = f_1|_{K^{(n-1)}}$ and h is the trivial homotopy, we simply write $c^n(f_0, f_1)$ for $c^n(f_0, f_1, h)$ and call it the **n -th difference cochain** of (f_0, f_1) . There are some basic facts about $c^n(f_0, f_1, h)$:

- h can be extended to a homotopy $H : K^{(n)} \times I \rightarrow Y$ between f_0 and f_1 if and only if $c^n(f_0, f_1, h) = 0$;
- (Naturality) Suppose L is another CW complex and $\phi : L \rightarrow K$ is a cellular map, then $c^n(f_0 \circ \phi, f_1 \circ \phi, h \circ (\phi \times \text{id}_I)) = \phi^\# c^n(f_0, f_1, h)$.
- $c^n(f_0, f_2, h * h') = c^n(f_0, f_1, h) + c^n(f_1, f_2, h')$, where $h * h'$ is the concatenation of h and h' . Especially, if $f_0|_{K^{(n-1)}} = f_1|_{K^{(n-1)}} = f_2|_{K^{(n-1)}}$ and h, h' are trivial homotopies, then $c^n(f_0, f_2) = c^n(f_0, f_1) + c^n(f_1, f_2)$.

In general, $c^n(f_0, f_1, h)$ is not a cocycle. But we have the following formula.

$$\textbf{Theorem 2.} \quad \delta c^n(f_0, f_1, h) = c^{n+1}(f_1) - c^{n+1}(f_0).$$

There is a useful lemma about difference cochains.

Lemma 3. *For any $f : K^{(n)} \rightarrow Y$ and any cochain $d \in C_{cell}^n(K; \pi_n(Y))$, there exists a map $f' : K^{(n)} \rightarrow Y$ such that $f'|_{K^{(n-1)}} = f|_{K^{(n-1)}}$ and $c^n(f, f') = d$.*

With the above preparations, we have the following theorem:

Theorem 4 (Eilenberg). *Given a map $f : K^{(n)} \rightarrow Y$, if and only if f can extend to $K^{(n+1)}$ after having been changed on $K^{(n)} \setminus K^{(n-1)}$.*

Proof. ($\Leftarrow$) : If $\exists f' : K^{(n+1)} \rightarrow Y$ such that $f'|_{K^{(n-1)}} = f|_{K^{(n-1)}}$, then

$$c^{n+1}(f) = c^{n+1}(f) - c^{n+1}(f') = \delta d^n(f', f)$$

Thus $c^{n+1}(f)$ is a coboundary.

($\Rightarrow$) : If $[c^{n+1}(f)] = 0$, then $\exists d \in C^n(K; \pi_n(Y))$ such that $c^{n+1}(f) = \delta d$. By lemma, there exists $f' : K^{(n)} \rightarrow Y$ such that $f|_{K^{(n-1)}} = f'|_{K^{(n-1)}}$ and $d^n(f, f') = -d$. So

$$c^{n+1}(f') = c^{n+1}(f) + \delta d^n(f, f') = \delta d - \delta d = 0.$$

Thus f' can be extended to $K^{(n+1)}$. □

Example 5. If the target space Y is n -connected, i.e., $\pi_k(Y) = 0$, $k = 0, 1, \dots, n$, then every map $f : K^{(0)} \rightarrow Y$ can be extended to $K^{(n+1)}$. Besides, any two maps $f, g : K^{(n)} \rightarrow Y$ are homotopic. Indeed, $f|_{K^{(0)}} \simeq g|_{K^{(0)}}$ for the reason that K is path-connected. We denote the homotopy by h_0 . Then h_0 can be extended to $h_n : K^n \times I \rightarrow Y$ since all the deformation cochain vanish.

Corollary 6. *If Y is n -connected and $\dim K \leq n$, then all the maps from K to Y are null-homotopic.*

Theorem 7 (Hopf-Whitehead). *and $(n-1)$ -connected space Y , there is a one-to-one correspondence:*

$$[K, Y] \longleftrightarrow H^n(K; \pi_n(Y)).$$

If we weaken the dimensional condition and instead strengthen the target space to be an Eilenberg-MacLane space $K(\pi, n)$. Then we will get another theorem:

Theorem 8. *For any CW complex X , group π and integer n , there is a one-to-one correspondence:*

$$[X, K(\pi, n)] \longleftrightarrow H^n(X; \pi).$$

Remark. In fact, there is a special class α in

$$H^n(K(\pi, n); \pi) \cong \text{Hom}(H_n(K(\pi, n)), \pi) \cong \text{Hom}(\pi, \pi)$$

which corresponds to the identity map in $\text{Hom}(\pi, \pi)$. The correspondence can be explicitly written by

$$\begin{aligned} [X, K(\pi, n)] &\rightarrow H^n(X; \pi) \\ f &\mapsto f^* \alpha. \end{aligned}$$

Characteristic Classes via Obstruction Theory

Cohomology with Local Coefficients

To describe characteristic classes as obstruction classes, we need cohomology with local coefficients.

Local Coefficient System:

Suppose X is a path-connected space. For any $x \in X$, let G_x be an abelian group attached to x . For any path γ_{xy} from x to y , we have an isomorphism $(\gamma_{xy})_* : G_x \rightarrow G_y$. If $(\gamma_{xy})_*$ only depends on the homotopy class of γ rel. endpoints and satisfies the following conditions:

$$(\text{constant path})_* = \text{Id}, \quad (\gamma_{xy} * \gamma_{yz})_* = (\gamma_{yz})_* \circ (\gamma_{xy})_*.$$

Then we call $\mathcal{G} = \{G_x\}_{x \in X}$ a **local coefficient system** on X . If we fix a base point $x_0 \in X$, then $\mathcal{G}$ is determined by the group $G = G_{x_0}$ and an action $\rho : \pi_1(X, x_0) \rightarrow \text{Aut}(G)$. We call G the **typical group** of $\mathcal{G}$.

Example 9. If $\mathcal{G}$ is a local coefficient system on X with typical group G and trivial action. Then it is the ordinary constant coefficient we are familiar with.

Example 10. If $G_x = \mathbb{Z}$ for all $x \in X$ and $\pi_1(X, x_0)$ acts on $\mathbb{Z}$ by multiplication of ± 1 according to whether the orientation is preserved or reversed along the loop. Then $\mathcal{G}$ is called the **orientation system** on X and denoted by Or .

Example 11. Let $p : \tilde{X} \rightarrow X$ be the universal covering of X . Then $\mathcal{G} = \{H_n(p^{-1}(x))\}_{x \in X}$ is a local coefficient system on X with typical group $H_n(\tilde{X})$ and action induced by the deck transformation group $\pi_1(X, x_0)$.

Example 12. Let $E \rightarrow X$ be a vector bundle with fiber F . Then $\mathcal{G} = \{H_n(p^{-1}(x))\}_{x \in X}$ is a local coefficient system on X with typical group $H_n(F)$ and action induced by the structure group of E .

Example 13. $\mathcal{G} = \{\pi_n(X, x)\}_{x \in X}$ is a local coefficient system on X with typical group $\pi_n(X, x_0)$ and action induced by the fundamental group $\pi_1(X, x_0)$.

Cohomology with Local Coefficients:

Let X be a CW complex and $\mathcal{G} = \{G_x\}_{x \in X}$ be a local coefficient system on X . We define the cellular cochain groups with local coefficients by

$$C_{\text{cell}}^n(X; \mathcal{G}) = \left\{ f : \{n\text{-cells of } X\} \rightarrow \bigsqcup_{x \in X} G_x \mid f(e_\alpha^n) \in G_{x_\alpha} \right\}$$

where x_α is a chosen point in the n -cell e_α^n . The coboundary operator $\delta : C_{cell}^n(X; \mathcal{G}) \rightarrow C_{cell}^{n+1}(X; \mathcal{G})$ is defined by

$$(\delta f)(e_\beta^{n+1}) = \sum_{\alpha} d_{\beta\alpha}(\gamma_{\alpha\beta})_* f(e_\alpha^n),$$

where $d_{\beta\alpha}$ is the degree of the map $\partial e_\beta^{n+1} \rightarrow X^{(n)}/(X^{(n)} - e_\alpha^n) \cong S^n$ and $\gamma_{\alpha\beta}$ is a path from x_α to x_β . It can be verified that $\delta^2 = 0$. Thus we can define the cohomology groups with local coefficients by

$$H^n(X; \mathcal{G}) = \frac{\ker(\delta : C_{cell}^n(X; \mathcal{G}) \rightarrow C_{cell}^{n+1}(X; \mathcal{G}))}{\text{im}(\delta : C_{cell}^{n-1}(X; \mathcal{G}) \rightarrow C_{cell}^n(X; \mathcal{G}))}.$$

The Homotopy Groups of $V_k(\mathbb{R}^n)$

Fact: The Stiefel manifold $V_k(\mathbb{R}^n)$ is $(n - k - 1)$ -connected and the first nontrivial homotopy group is

$$\pi_{n-k}(V_k(\mathbb{R}^n)) = \begin{cases} \mathbb{Z}, & n - k \text{ is odd or } k = 1; \\ \mathbb{Z}_2, & n - k \text{ is even.} \end{cases}$$

Characteristic Classes

Now let's back to vector bundles. Let ξ be an $\mathbb{R}^n$ -bundle over a CW complex B . We want to investigate how nontrivial ξ is. To do this, we try to find k -linearly independent sections of ξ . First suppose there is a Euclidean metric g on ξ . Then we can talk about orthonormal sections.

Remark. Any real vector bundle over a paracompact space admits a Euclidean metric. Each two Euclidean metrics g_0, g_1 are homotopic through $g_t = (1 - t)g_0 + tg_1$.

Let $V_k(\xi)$ be the Stiefel manifold bundle associated to ξ . It's a fiber bundle over B with fiber $F = V_k(\mathbb{R}^n)$. A section s of $V_k(\xi)$ is a k -orthonormal frame of ξ . Locally on a trivializing neighborhood U_α , s can be identified as a map from U_α to $V_k(\mathbb{R}^n)$. Thus we can use obstruction theory to study the existence of sections of $V_k(\xi)$.

The obstructions $c^{j+1}(s)$ lie in $H^{j+1}(B; \mathcal{G})$, where $\mathcal{G}$ is a local coefficient system on B with typical group $\pi_j(V_k(\mathbb{R}^n))$. This is because although the restriction of $V_k(\xi)$ on every j -cell e_α^j is trivial, the transition function may not be identity, from which induces a nontrivial isomorphism of homotopy

groups. Moreover, any path γ_{xy} from x to y can lift to a homotopy from F_x to F_y , thus induces an isomorphism

$$(\gamma_{xy})_* : \pi_j(F_x, x) \xrightarrow{\cong} \pi_j(F_y, y).$$

The theorems about obstruction classes can all generalize to cohomology with local coefficients without difficulty.

Now since $V_k(\mathbb{R}^n)$ is $(n-k-1)$ -connected, all the section s can be extended to $B^{(n-k)}$. The first obstruction comes when extend s to $(n-k+1)$ -skeleton, which we denote by

$$\mathfrak{o}_{n-k+1} := c^{n-k+1}(s) \in H^{n-k+1}(B; \{\pi_{n-k}(V_k(\mathbb{R}^n))\})$$

Since all sections on $B^{(n-k)}$ are homotopic with each other, this obstruction class does not depend on the choice of s , i.e, $\mathfrak{o}_{n-k+1}$ it is well-defined.

When $n-k = 2j-1$ is odd and $k \neq 1$, $\pi_{n-k}(V_k(\mathbb{R}^n)) = \mathbb{Z}_2$. In this situation the local coefficient system is trivial. Thus the cohomology is in fact the ordinary cohomology $H^{2j}(B; \mathbb{Z}_2)$.